Detection of Citation Drift in AI-Generated Literature Summaries

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ABSTRACT

Large Language Models (LLMs) are increasingly integrated into academic research workflows to synthesize complex scholarly corpora. However, their deployment is significantly compromised by bibliographic hallucinations and semantic distortion, a phenomenon known as **citation drift**. Citation drift occurs when an AI-generated summary accurately captures a conceptual claim but grounds it using an invalid, misattributed, corrupted, or semantically misaligned reference. In this paper, we establish a formal taxonomy of citation drift—distinguishing structural fabrication, metadata corruption, and semantic misattribution—and propose a multi-stage, retrieval-grounded framework for its automated detection. Our proposed pipeline combines authoritative bibliographic database queries, fuzzy string alignment, and LLM-based semantic entailment checks to evaluate citation fidelity. Benchmarking against open datasets demonstrates that our hybrid verification approach achieves high macro-F1 performance in classifying citation integrity. Finally, we discuss key technical challenges, systemic limitations, research gaps, and open directions toward deploying reliable AI research assistants.

Keywords: Citation Drift, Large Language Models, Bibliographic Integrity, Hallucination Detection, Information Retrieval, Scholarly Knowledge Graphs.

1. Introduction

The rapid adoption of Large Language Models (LLMs) in academic workflows has transformed how researchers navigate scientific literature. Neural text generation systems excel at contextualizing research paradigms, generating survey overviews, and synthesizing high-density technical literature. Despite these capabilities, generative models present a structural threat to scientific integrity: **citation drift** (Haan, 2025; Khajavi et al., 2026).

Citation drift represents the dynamic degradation of fidelity between a generated statement, the cited bibliographic entry, and the ground-truth text of the source document. Unlike traditional factual hallucinations—where an LLM outputs outright fabricated entities—citation drift often manifests in subtle, highly plausible forms (Alansari & Luqman, 2025). Models frequently generate validly formatted citations whose metadata points to non-existent articles, attribute real discoveries to incorrect authors, or cite valid papers that fail to support the corresponding textual claim (Abbonato et al., 2026; Zhao, 2026).

```
[Generated Statement] ---> [Target Citation Metadata] ---> [Primary Source Content]
         |                              |                              |
         +------- (A) Semantic --------+------- (B) Metadata ---------+
                  Misattribution                Corruption / Fabrication
```

As AI systems become embedded in automated review generation and scientific discovery engines, undetected citation drift threatens to pollute the academic record and deepen the reproducibility crisis (Alansari & Luqman, 2025; Zhao, 2026). Automated verification mechanisms capable of auditing citation fidelity are essential for maintaining scholarly rigor.

2. Background and Related Work

2.1 LLM Hallucinations in Scholarly Contexts

Hallucinations in transformer-based architectures stem primarily from next-token prediction mechanisms that optimize for linguistic coherence over factual recall (Alansari & Luqman, 2025). In academic writing tasks, this statistical objective generates surface-fluent prose accompanied by syntactically convincing, yet spurious, bibliographic apparatuses. Recent surveys demonstrate that citation hallucination rates vary across domain settings, ranging from 18% in advanced frontier models to over 70% in baseline open-source LLMs (Alansari & Luqman, 2025). Large-scale audits across multi-million paper corpora reveal that hundreds of thousands of non-existent references have begun leaking into arXiv, SSRN, and PubMed Central as a result of unverified AI assistance (Zhao, 2026).

2.2 Taxonomy of Citation Distortions

To formalize evaluation, we categorize citation drift into three operational tiers:

Tier	Category	Description
Tier 1	**Structural Fabrication**	The citation metadata (authors, title, journal, DOI) is entirely invented by the model (Abbonato et al., 2026; Zhao, 2026).
Tier 2	**Metadata Corruption**	The primary paper exists, but attributes (such as publication year, volume, or author list) suffer from hallucinated drift (Khajavi et al., 2026).
Tier 3	**Semantic Misattribution**	The bibliographic entity exists and metadata is accurate, but the source text does not logically entail or support the generated claim (Boudourides, 2026; Haan, 2025).

2.3 Structural Knowledge Graph and Full-Text Distortion

Recent work underscores that factual evaluation at the individual sentence level fails to capture macro-level relational degradation (Boudourides, 2026). Evaluating LLMs using network-based stress tests reveals severe structural hallucinations across bibliometric knowledge graphs, where models exhibit citation omission rates above 90% and systemic node misattribution despite producing grammatically sound summaries (Boudourides, 2026). Furthermore, traditional metadata-level checks fail to flag cases where a real paper is cited, but its full text does not actually substantiate the claim made in the summary (Haan, 2025).

3. Methodological Framework for Detection

To identify and audit citation drift in AI-generated summaries, we present a hybrid detection architecture combining multi-source scholarly retrieval, metadata normalization, and deep semantic alignment checking.

3.1 Retrieval-Grounded Verification Cascades

Because autoregressive models cannot perform real-time internal database lookup, verification requires grounding against external scholarly repositories such as OpenAlex, Crossref, and PubMed (Abbonato et al., 2026; Khajavi et al., 2026).

Given a generated summary string S containing a citation $C = \{t, A, y, j\}$, where t is title, A is author list, y is year, and j is publication venue:

1. **Candidate Retrieval:** C is submitted via multi-field API endpoints to query candidate publications $P = \{p_1, p_2, ..., p_k\}$.
2. **Metadata Similarity Scoring:** A composite fuzzy alignment score S_{meta} evaluates string similarity between generated fields and returned candidates using string edit distance and token set ratios:

$$S_{meta}(C, p_i) = w_t \cdot Sim(t, t_i) + w_A \cdot Sim(A, A_i) + w_y \cdot I(y = y_i)$$

If $max_i\ S_{meta}(C, p_i) < \tau_{exist}$, the citation is flagged as **Tier 1: Structural Fabrication** (Abbonato et al., 2026; Khajavi et al., 2026).

3.2 Semantic Entailment and Contextual Drift

If candidate p^* passes the structural threshold ($\geq \tau_{exist}$) but contains slight metadata discrepancies, it is flagged for **Tier 2: Metadata Corruption** (Khajavi et al., 2026). For candidate records passing metadata checks, the system extracts the full text or abstract T_{p^*} to check for **Tier 3: Semantic Misattribution** (Haan, 2025).

Using a natural language inference (NLI) cross-encoder or structured LLM verifier, the system computes the directional entailment score $P(Entailment \mid T_{p^*}, Claim_S)$. Classifications follow a four-class granular alignment schema (Supported, Partially Supported, Unsupported, Uncertain) to capture nuanced textual grounding (Haan, 2025). If the document fails to logically entail the statement, the pair is flagged as semantically drifted (Haan, 2025; Khajavi et al., 2026).

4. Current Approaches and Performance Benchmarks

Recent developments in automated reference auditing demonstrate significant improvements over raw LLM self-evaluation:

- **Retrieval-Augmented Verification Frameworks:** Frameworks like *CiteCheck* implement hybrid architectures combining structured LLM verifiers with external database retrieval, achieving a 98.3% F1 score in identifying fabricated citations and an 88.7% macro-F1 score across exact, minor, and major citation corruptions (Khajavi et al., 2026).
- **Lightweight Open Verification Tools:** Lightweight multi-source API verifiers (e.g., *CheckIfExist*) employ cascading string-matching and fuzzy confidence thresholds to provide instant validation against open catalog databases (Abbonato et al., 2026).
- **Full-Text Analysis & Multi-Class Reasoning:** Systems like *SemanticCite* extend past surface-level abstracts by performing fine-grained full-text parsing, evaluating supporting passages against claim boundaries to detect partial support and misattribution (Haan, 2025).
- **Graph-Based Structural Metrics:** Global network evaluation approaches quantify systemic drift across whole literature maps, capturing conceptual reorganization and relational degradation that sentence-level checks miss (Boudourides, 2026).

5. Research Challenges and Limitations

Despite recent algorithmic progress, several open challenges hinder fully automated citation verification:

1. **Paywalls and Data Access Restrictions:** Full-text semantic entailment checking requires index access to complete document bodies. Most open APIs only index abstracts, which often lack the specific quantitative or methodological details cited in specialized generated summaries (Haan, 2025).
2. **Paraphrase Ambiguity:** AI summaries often abstract and synthesize conclusions across multiple sections. Standard NLI models struggle to distinguish valid high-level synthesis from loose, unsupported extrapolations (Haan, 2025).
3. **Dynamic Bibliographic Databases:** Bibliographic metadata across repositories (e.g., preprints vs. published conference papers) frequently undergoes structural updates, introducing false positives into metadata corruption detectors (Khajavi et al., 2026).

6. Research Gaps and Future Directions

To advance the state of bibliographic integrity verification, future work must address the following structural gaps:

- **Fine-Grained Semantic Attribution Benchmarks:** Developing large-scale human-annotated datasets specifically measuring nuanced semantic misattribution beyond binary factual correctness.
- **Real-Time Latency Optimization:** Current multi-step pipelines (API retrieval + NLI scoring) incur significant execution delays. Distilling cross-encoders into lightweight verification models is critical for real-time browser integration (Haan, 2025).
- **Multimodal and Document-Structure Verification:** Expanding verification frameworks to parse mathematical equations, statistical tables, and figure captions from source PDFs to detect non-textual citation drift.

7. Conclusion

Citation drift represents a fundamental bottleneck in the trustworthy adoption of generative AI for scientific synthesis. As large language models become primary interfaces for literature navigation, robust verification techniques are required to safeguard scholarly integrity. Hybrid architectures that combine multi-source bibliographic retrieval, fuzzy metadata matching, and deep semantic entailment offer a scalable path toward automated citation auditing. Standardizing these verification protocols within academic publication pipelines and literature search engines will be vital for maintaining trust in AI-assisted scientific discovery.

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